

Richard Xu

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EDUCATION

Western University

London, ON

Bachelor of Engineering Science (BESc), Software Engineering — GPA: 3.9/4.0

Sept. 2024 – Apr. 2028

Ivey Business School, Honours Business Administration (HBA) Dual Degree Candidate

Sept. 2027 – Apr. 2029

EXPERIENCE

AI/Software Engineering Intern

May 2026 – Present

Magical

Toronto, ON

- Built production healthcare-claims automations using **Node.js, Typescript, Express, REST APIs, Redis/Upstash, and browser automation**, translating operations requirements into EHR and CRM workflows integrated with Gmail and Slack
- Accelerated full-dataset CSV exports by approximately **20×**, reducing runtime from **5.5 minutes to 17 seconds** by replacing nearly 13,000 per-record Redis requests with batched HMGET pipelines across 6,499 records
- Built a centralized **Slack-based error-alerting system** for claims workflows, auditing **150 production runs** to identify **7 incomplete executions** and surfacing agent-level failure context with direct run links for faster triage
- Eliminated a **6.8% reporting reconciliation gap** through cohort-based deduplication and validation across six claim statuses, supporting accurate daily and monthly reporting for **1,600+ claims**
- Hardened production workflows by replacing brittle nested JSON parsing with defensive API handling for escaped, double-encoded, empty, and null payloads, then recovered structured CRM data from **8 failed runs**

Software Engineering Intern

Jan. 2026 – April. 2026

Autumn (Startup)

Remote - New York, NY

- Built an MVP for a full-stack pricing and workflow platform using **Next.js, Strapi/Node.js, and PostgreSQL**, reducing manual service configuration for funeral-service providers
- Designed backend APIs and data models for dynamic pricing, service configurations, and structured records covering more than **15,000 U.S. funeral homes**

AI/ML Research Intern

May 2025 – Aug. 2025

Western University

London, ON

- Built a Wi-Fi optimization platform with a **PPO reinforcement-learning agent** using **Stable-Baselines3 and Gymnasium**, dynamically switching access points to optimize latency, jitter, and packet loss
- Engineered a **Flask and React** experimentation platform and benchmarked the agent against static and random strategies across **50+ reproducible episodes per run**

PROJECTS

Slack AI Digester | *JavaScript, Node.js, Express, Claude API, REST APIs*

Apr. 2026

- Built a role-aware digest prototype that converts Slack-style engineering updates into phase-aware summaries for **multi team personas**
- Designed modular REST endpoints and parallel generation with **Promise.allSettled()**, isolating failures so one persona request did not block remaining outputs

eBay Deal Finder | *Python, Django, HTML/CSS, eBay Browse API*

Jul. 2025 – Aug. 2025

- Built a Django application that queries live listings across U.S., Canadian, and U.K. eBay marketplaces and ranks results by price or review quality
- Implemented secure API credential handling, dynamic search results, and direct purchase links to streamline cross-marketplace product comparison

TECHNICAL SKILLS

Languages: Python, Java, Javascript, HTML/CSS, C#, Typescript

AI / ML: LLMs (API-based), Prompt Engineering, RAG, Agentic Systems, Reinforcement Learning

Frameworks: React, Next.js, Flask, Django, FastAPI, Strapi (Node.js), .NET, Unity, LangChain, LangGraph

Libraries/Databases: pandas, NumPy, Matplotlib, PostgreSQL, Redis/Upstash